

SAMPLE SYLLABUS: INTRODUCTION TO POLITICAL SCIENCE METHODS

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Course Description and Objectives

The primary objective of this course is to provide students with an introduction to methods relevant to answering rigorous questions in the field of political science. This course covers both quantitative and qualitative methods. The first half of the course will teach the basics of causal inference, with lectures on topics such as potential outcomes, confounders, directed acyclic graphs, and linear regressions. Interactive in-class activities and weekly problem sets that use R/R Studio will be paired with the topics for students to apply their knowledge. The second half of the course addresses qualitative approaches such as interviews, archives, and process tracing. The final project asks students to answer a political science question of their choice using either a quantitative or qualitative approach in a research report.

Course Format

This is an introductory course consisting of lectures, interactive modules, a midterm exam, and a final project.

Course Structure and Assignments

Preparation for each class, as well as the final research project are important for this course. The assignments and work requirements for the students are as follows:

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| (1) Class participation | 10% |
| <ul style="list-style-type: none">• Attend every session having any required readings or pre-class lectures digested.• Repeated tardies or absence without legitimate excuse will negatively affect this grade. | |
| (2) Midterm exam | 30% |
| <ul style="list-style-type: none">• Students will write an in-class exam based on the course content in the first half of the course. The content of the exam will reflect the problem sets. A study guide will be given to students at least two weeks before the exam. | |
| (3) Problem sets (5) | 25% |
| <ul style="list-style-type: none">• Problem sets will be interspersed throughout the course that reflect lecture/class content.• Students are free and encouraged to work on these with each other; however, the work that students turn in must be their own.• Use of AI MUST be disclosed up front and is limited to coding help (conceptual help not allowed). If the submitted work is found to be AI-generated in any capacity without prior disclosure, the problem set will receive a zero and the student will be referred to the Office of Academic Misconduct, without questions. | |

(4) Final Presentations

5%

- Each student will be asked to give an 6-8 minute presentation on their final research paper during the last few weeks of class.
- Students will be graded on the delivery/clarity of their oral presentation, the content/organization of the presentation itself, and the questions each student asks their peers during others' presentations.

(5) Final Research paper

30%

- Students can choose a question/topic that utilizes methods learned in the course to answer. The paper must be about 15-20 double-spaced pages (approximately 5,000 words, no more than 6,000). Any paper that does not follow these rules will be penalized, unless the instructor specifically gives you written permission.
- Collaboration is important in Political Science and encouraged. Students may partner up for the final research report if they wish to do so.
- Students **MUST** schedule at least one office hour session before Class Meeting 20 to talk about their paper idea with the instructor. Office hours after that can be scheduled at any time, but any report submitted without any office hours will be subject to grade deductions.

Assignment Submission Policy

- You will be given a total of 24 “late” hours to use throughout the semester. For example, if a problem set is due at 11:59 and you submit it at 8 AM, you use 8 hours of your total pool, with no late penalty.
- If you use up your 24 hours and submit more late assignments, each assignment will be deducted by half a letter grade each day it is late.
- At the end of the semester, any unused late hours will be converted to extra credit. Each hour accounts to 0.1%, so 24 unused hours could get you a 2.4% boost in your grade at the end of the semester.
- Requested extensions **accompanied with doctor’s notes, grievance notes**, etc will be exempt from the late policy.

Course Schedule (based on a semester schedule, meeting twice a week)

REQUIRED TEXTBOOKS:

Paul M.Kellstedt & Guy D. Whitten (2009). The Fundamentals of Political Science Research, 3rd ed. Cambridge: Cambridge University Press. – Referred to in the syllabus as “K&W”

Class Meeting 1 Explanations of the course goals and syllabus

Class Meeting 2 Introduction to Causality and Validity

Henry Brady (2008). "Causation and Explanation in Social Science." in The Oxford Handbook \ of Political Methodology.

Donald T. Campbell & Julian Stanley. (1963). "Experimental and Quasi-Experimental Designs for Research." (pp. 1-6)

Class Meeting 3 and 4 Causality continued: Fundamental Problem of Causal Inference, Confounders, Failures

K&W 3.1-3.3

Jana von Stein (2005). "Do Treaties Constrain or Screen? Selection Bias and Treaty Compliance." *American Political Science Review*, 99(4): 611-622.

PROBLEM SET 1 DUE AT 11:59 SUNDAY

Class Meeting 5 and 6 Conceptualization and Measurement

K&W Ch. 5 & 6.2

Robert Adcock & David Collier (2001). "Measurement Validity: A Shared Standard for Qualitative and Quantitative Research." *American Political Science Review*, 95(3): 529-546.

Class Meeting 7 and 8 DAGs and Causal Relationships

Digitale, J. C., Martin, J. N., & Glymour, M. M. (2022). Tutorial on directed acyclic graphs. *Journal of clinical epidemiology*, 142, 264–267.

PROBLEM SET 2 DUE AT 11:59 PM ON SUNDAY

Class Meeting 9 and 10 From Concepts to Data

Hakeem Jefferson. (2019). "The Curious Case of Black Conservatives."

Amanda Marcotte (2014). "Best Way for Professors to Get Good Student Evaluations? Be Male." <https://slate.com/human-interest/2014/12/gender-bias-in-student-evaluations-professorsof-online-courses-who-present-as-male-get-better-marks.html>

Read this: [Types of Sampling](#)

Do this: <http://rocknpoll.graphics>

Class Meeting 11 and 12 Evaluating and Describing Data

K&W: 6.1, 6.3, 6.5

BRING COMPUTER WITH R INSTALLED. <https://cran.r-project.org/>

If R is not installed on your computer by class, your participation grade will be affected.

Class Meeting 13 and 14 More data work! Regressions and relationships between variables

The Cartoon Guide to Statistics: pp. 19-26

K&W 9.1-9.3

PROBLEM SET 3 DUE AT 11:59 PM ON SUNDAY

Class Meeting 15 MIDTERM REVIEW

Class Meeting 16 MIDTERM

Class Meeting 17 Special Module: Forming a good research question

Optional reading:

Gustafsson, K., & Hagstrom, L. (2023). The insecurity of doing research and the ‘so what question’ in political science: How to develop more compelling research problems by facing anxiety. *European Political Science*, 15.

Class Meeting 18 and 19 Hypothesis Testing and Confidence Intervals

Watch pre-filmed lecture, and come to class with questions + ready to do in-class lab work

Class Meeting 20 Pitfalls of Quantitative Research

Christopher Achen (2005). “Let’s Put Garbage-Can Regressions and Garbage-Can Probits Where They Belong.” *Conflict Management and Peace Science*, 22(4): 327-339.

PROBLEM SET #4 DUE 11:59 ON SUNDAY

Class Meeting 21 Intro to Qualitative Research: Case studies

Sidney Tarrow (2010). "The Strategy of Paired Comparison: Toward a Theory of Practice." *Comparative Political Studies*, 43(2): 230-259.

David A. Freedman (1991). "Statistical Models and Shoe Leather." *Sociological Methodology*, 21(2): 291-313.

Mukhija, Vinit. 2010. "N of One plus Some: An Alternative Strategy for Conducting Single Case Research." *Journal of Planning Education and Research* 29 (4): 416–26.

Class Meeting 22 and 23 Qualitative Research, cont: Process Tracing

Collier, David. 2011. "Understanding Process Tracing." *PS: Political Science and Politics* 44 (4): 823–30.

Zaks, Sherry. 2018. "Relationships Among Rivals: A Framework for Analyzing Contending Hypotheses in Process Tracing." *Political Analysis* 25 (3): 344–62.

PROBLEM SET #5 DUE 11:59 ON SUNDAY

Class Meeting 24 and 25 Interviewing

Fujii, Lee Ann. 2018. "What Is Relational Interviewing?" In *Interviewing in Social Science Research: A Relational Approach*. Routledge.

Lamont, Michele, and Ann Swidler. 2004. "Methodological Pluralism and the Possibilities and Limits of Interviewing." *Qualitative Sociology* 37 (2): 153–71.

Class Meeting 26 Final Paper Presentations